ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka. Approved by AICTE, New Delhi

Computer Science & Engineering (Data Science)

PROJECT

ON

From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics

Subject Code: BAD601

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TASK 1 : Big Data in Daily Life – Visual Storytelling

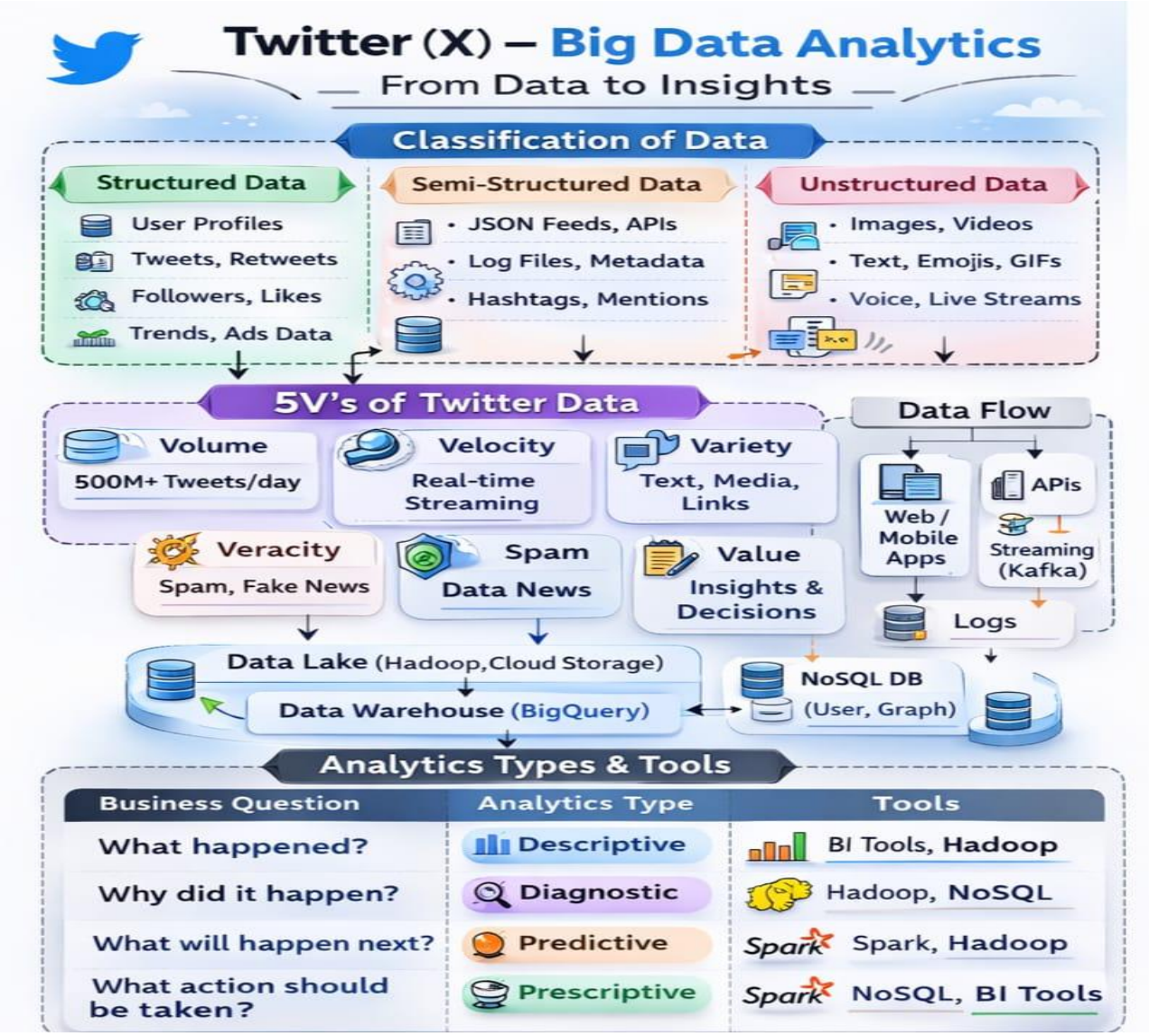

Fig 1.1 – Infographic Poster 1

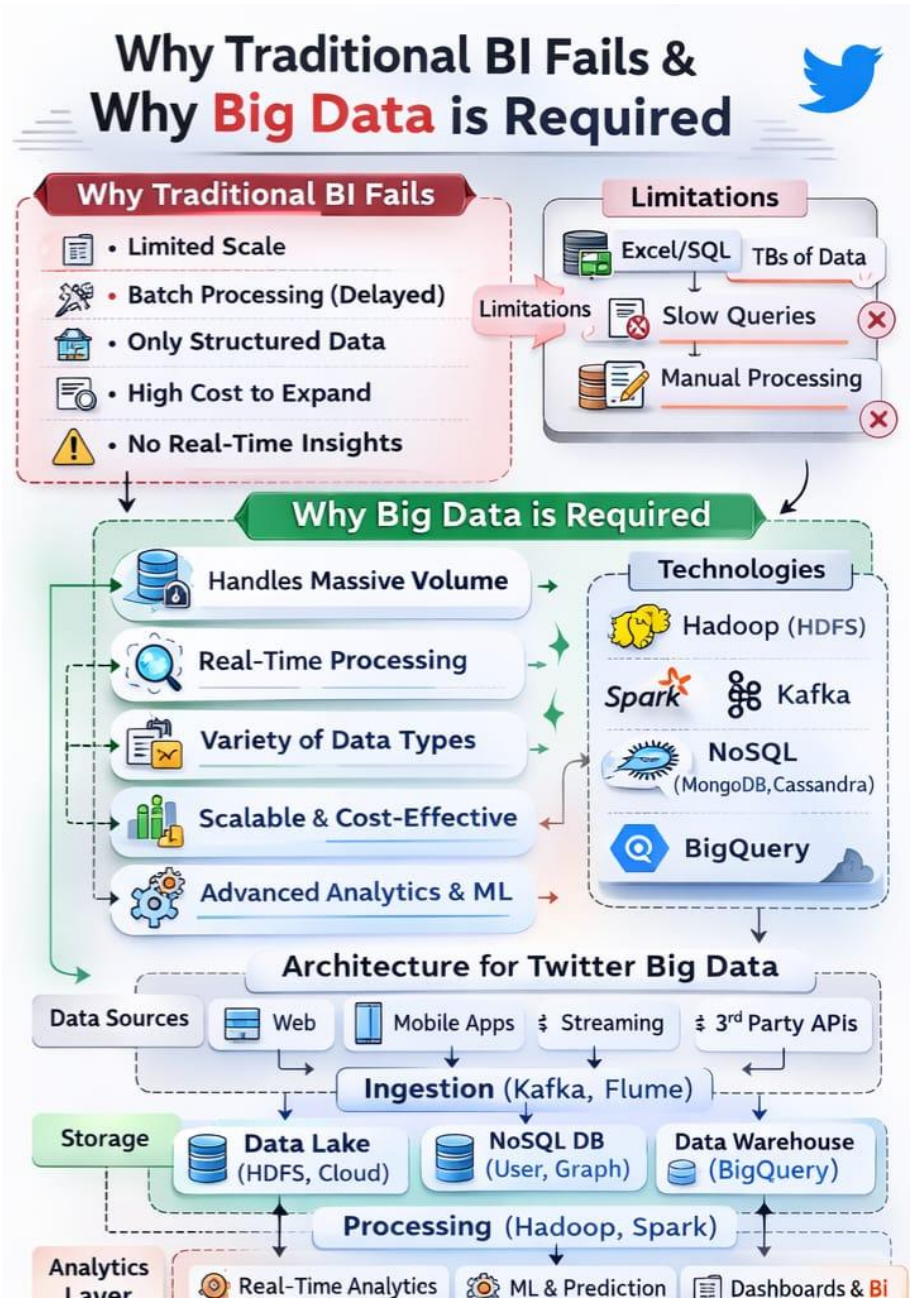

Fig 1.2 – Infographic Poster 2

TASK 2: BI VS BIG DATA – ROLE PLAY

Dialogue: BI vs Big Data

Manager: Hey, we're already using Excel and SQL for reports. They work fine. Why complicate things with Big Data?

Consultant: They work — but only up to a point. Tell me, how long does it take to prepare your monthly dashboard?

Manager: Two days minimum. Queries run overnight, then Excel cleanup and pivot tables.

Consultant: That's a classic traditional BI setup. The limitation is batch processing. You're always looking at yesterday's data, not what's happening right now.

Manager: But we don't need real-time data, do we?

Consultant: Let's take Spotify as an example. When a user skips a song, their recommendation engine updates almost instantly. That's only possible because they use streaming systems like Apache Kafka.

Manager: But we're not a streaming company.

Consultant: True. But customer expectations are shaped by companies like Spotify and Amazon. People expect personalization instantly — not next week.

Manager: So what exactly is the limitation of traditional BI?

Consultant: Four major ones:

1. It handles only structured data (tables with rows and columns).
2. Scaling means buying bigger, expensive servers.
3. It processes data in batches.
4. It struggles with huge volumes.

Manager: Okay, explain the evolution part.

Consultant: As businesses moved online, data exploded — website clicks, app logs, social media, IoT sensors. Traditional relational databases couldn't handle the "3Vs" — Volume, Velocity, and Variety. That's when Big Data technologies emerged.

Manager: Like Hadoop?

Consultant: Exactly. Apache Hadoop stores and processes massive data across clusters of computers. Instead of one powerful machine, it distributes work across hundreds of smaller ones.

Manager: Why not just upgrade SQL servers?

Consultant: That's vertical scaling — costly and limited. Hadoop uses horizontal scaling — just add more nodes. Much cheaper and flexible.

Manager: And NoSQL? We've always used relational databases.

Consultant: Relational databases require fixed schemas. But modern data changes constantly. NoSQL databases like MongoDB allow flexible, schema-less storage. Perfect for user activity logs, product reviews, or IoT data.

Manager: So Big Data is mainly about size?

Consultant: Not just size. It's also about speed and insight. Tools like Apache Spark process large datasets in memory for real-time analytics and machine learning.

Manager: Give me a real business benefit.

Consultant: Sure:

- Real-time fraud detection in banks.
- Dynamic pricing in airlines.
- Personalized recommendations in e-commerce.
- Predictive maintenance in manufacturing.
-

Manager: So traditional BI tells us what happened, and Big Data predicts what will happen?

Consultant: Exactly. BI is descriptive. Big Data enables predictive and prescriptive analytics.

Manager: Does that mean we throw away BI completely

Consultant: Not at all. Most companies adopt a hybrid approach. Keep SQL data warehouses for structured reporting. Add Big Data tools for real-time analytics and unstructured data.

Manager: And cloud makes this affordable?

Consultant: Yes. Platforms like Google Cloud or Amazon Web Services provide managed Hadoop and Big Data services. You pay only for usage.

Manager: So the real benefit is better decisions, faster action, and competitive advantage?

Consultant: Exactly. Traditional BI supports operations. Big Data drives innovation and growth.

Manager: I get it now. Excel and SQL are enough for reporting. But if we want real-time personalization, scalability, and predictive analytics — we need Big Data.

Consultant: That's the evolution. Not replacement — expansion. And the businesses that adapt early gain the edge.

TASK 3 : Architecture Design

Traditional Data Warehouse:

A Traditional Data Warehouse architecture is a centralized data management system designed to store and analyse structured data collected from multiple operational sources. It primarily supports Business Intelligence (BI) activities such as reporting, dashboard creation, and historical data analysis. The architecture mainly supports descriptive analytics and batch processing. However, it has limitations in handling unstructured data, real-time streaming data, and large-scale distributed workloads.

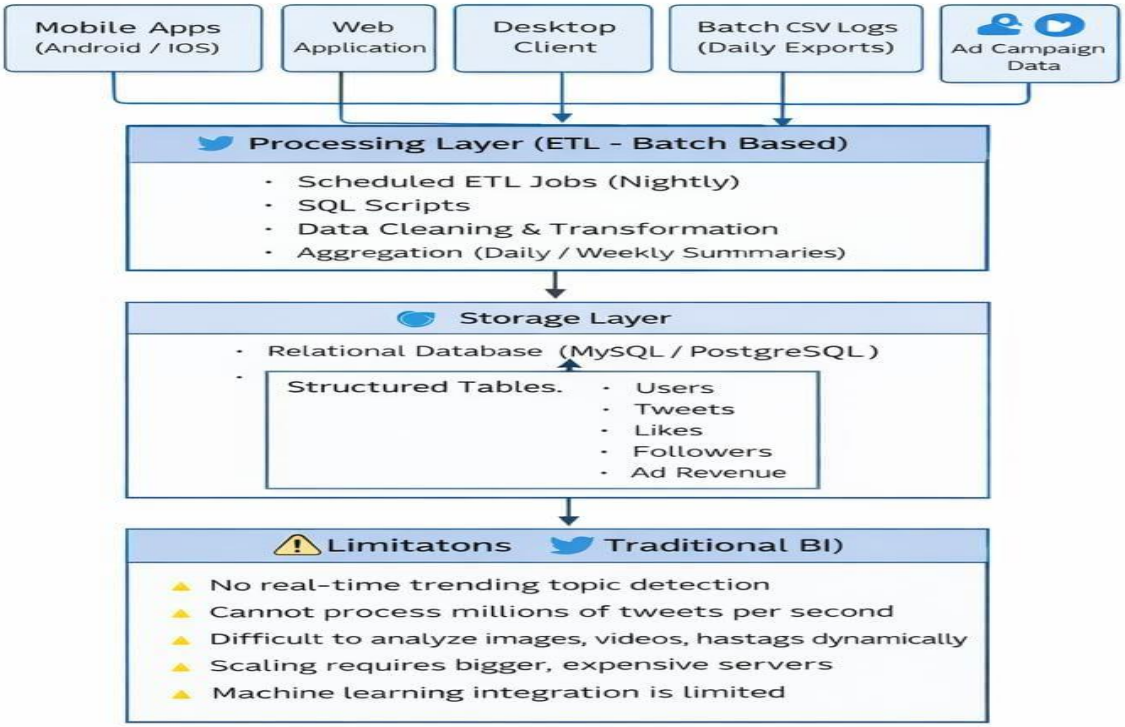

Fig 3.1 – Traditional Data Warehouse

Hadoop-based Big Data Architecture

Hadoop-Based Big Data Architecture is a distributed data processing framework designed to handle massive volumes of structured, semi-structured, and unstructured data. Unlike traditional centralized systems, this architecture distributes storage and computation across multiple nodes to ensure scalability and fault tolerance. This architecture enables horizontal scaling, high performance, and real-time data-driven decision-making.

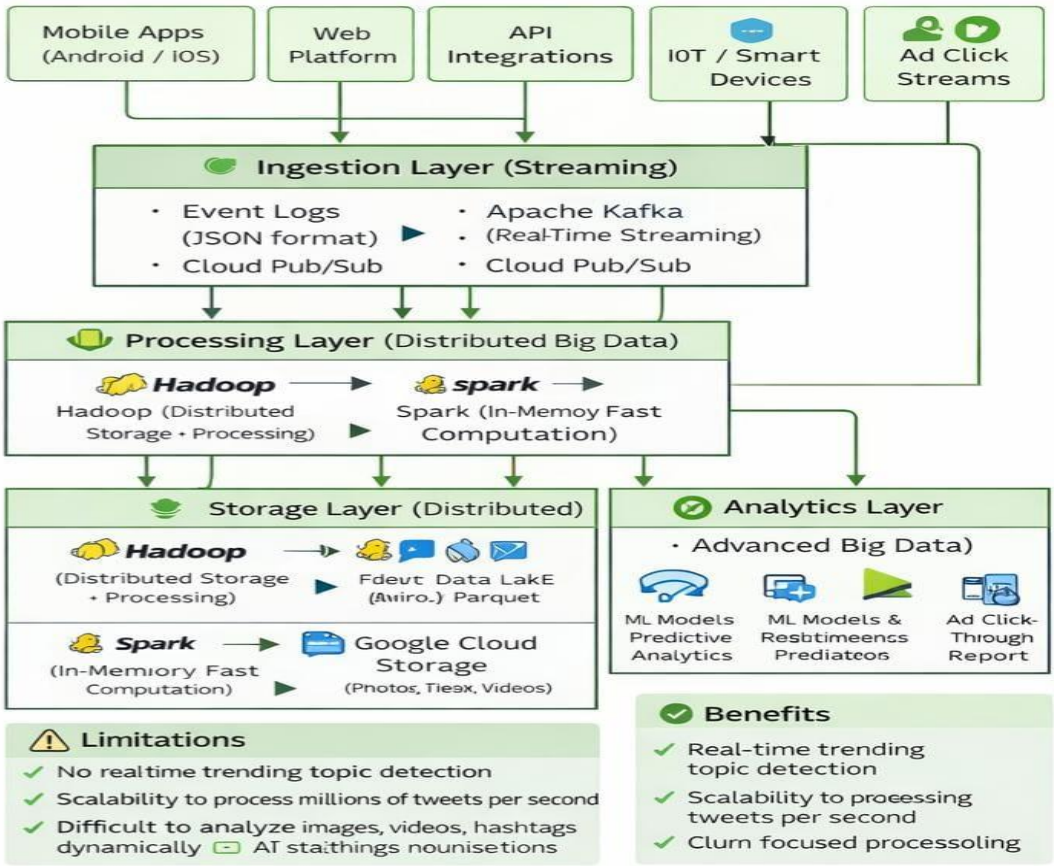

Fig 3.2 – Hadoop Big Data

TASK 4 : Understand the 4 Types of Analytics

Descriptive Analytics

Answers: What happened?

- Daily active users.
- Number of tweets posted
- Trending hashtags
- Tableau dashboards, Power BI reports, Hadoop are used for large log aggregation.

Diagnostic Analytics

Answers: Why did it happen?

- Why did a hashtag suddenly trend?
- Why did ad revenue decline?.
- Hadoop for log analysis and NoSQL (MongoDB/Cassandra) are used for event investigation

Predictive Analytics

Answers: What will happen next?

- Will a tweet go viral?.
- Will a hashtag trend tomorrow?
- Tools used are Apache Spark (ML models), Hadoop for large dataset training

Prescriptive Analytics

Answers: What action should be taken?

- Which tweets to promote?
- Which users to target with ads?
- Tools used are Spark ML + NoSQL + BI tools for decision dashboards.

Business Question	Analytics Type	Tool Used
What happened?	Descriptive Analytics	BI Tools (Tableau, Power BI), Hadoop
Why did it happen?	Diagnostic Analytics	Hadoo[p, NoSQL
What will happen next?	Predictive Analytics	Spark, Hadoop
What action should be taken?	Prescriptive Analytics	Spark, NoSQL, BI Tools

Table 4.1 – Analytics Table

BONUS CHALLENGE : Explain Big Data to a 10-Year-Old

Imagine **X** (Twitter) is like a huge world notice board

Millions and millions of people are writing on this board at the same time.
Every second, people are:
• Posting tweets
• Liking posts
• Retweeting
• Replying to others
• Following new accounts
• Sharing videos and news

Now imagine there is only one person reading everything and deciding:
• Who likes sports
• Who follows politics
• Who enjoys memes
• Who talks about technology

Do you think one person can read millions of tweets every second?
Impossible! They would get overwhelmed.
So instead of one person, Twitter uses many powerful computers working together
Each computer has a special job:
• One tracks what topics you like
• One notices which tweets you stop to read
• One finds trending hashtags
• One checks for spam accounts
• One suggests people you may want to follow
All these computers communicate fast.
Together, they act like a giant smart brain.
This brain can:

- Read millions of tweets every second
- Discover what is trending worldwide
- Show you posts you care about
- Keep the platform safe

This teamwork of computers handling huge amounts of information is called Big Data.
So, when Twitter shows you trending topics or tweets that match your interests…
It's not magic.
It's Big Data working behind the scenes.